WORKED SOLUTION

STEP II 1998, Question 13

Probability and Statistics • Exponential Distribution

STEP II 1998, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s2-q13

The key idea

The first result is the *memoryless property* of the exponential distribution: if a service has already lasted time t, the time still to go has the same distribution as a brand-new service. Once you have it, the shop story barely needs any calculation. Customers who are part-way through being served are, in terms of probability, just like customers starting afresh. So at every moment that matters, the race between the people still being served is a race between identical, independent exponential times.

Solution

The memoryless property

For $x \geq 0$,

$$\mathrm{P}(X > x) = \int_x^\infty \lambda \mathrm{e}^{-\lambda u}\,\mathrm{d}u = \Big[-\mathrm{e}^{-\lambda u}\Big]_x^\infty = \mathrm{e}^{-\lambda x}.$$

Take $s, t \geq 0$. The event $\{X > s+t\}$ is contained in $\{X > t\}$, so their intersection is just $\{X > s+t\}$. By the definition of conditional probability,

$$\begin{aligned}\mathrm{P}(X > s+t \mid X > t) &= \frac{\mathrm{P}(X > s+t \text{ and } X > t)}{\mathrm{P}(X > t)} = \frac{\mathrm{P}(X > s+t)}{\mathrm{P}(X > t)} \\ &= \frac{\mathrm{e}^{-\lambda(s+t)}}{\mathrm{e}^{-\lambda t}} = \mathrm{e}^{-\lambda s} = \mathrm{P}(X > s),\end{aligned}$$

which is the required result. (We could divide because $\mathrm{P}(X > t) = \mathrm{e}^{-\lambda t} > 0$.)

Both customers still being served

Model: the two service times are independent and each has the given exponential distribution. Let A and B be the two customers being served when I arrive.

Ask yourself Customer A has already been served for some unknown length of time. Doesn't that change the distribution of how much longer it takes?

No, and this is exactly what the first part tells us. Suppose A's service has lasted u so far. Then the probability that it lasts at least a further t is

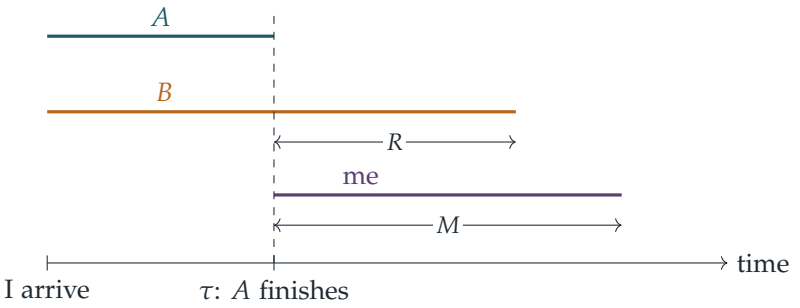
$$\mathrm{P}(X > u + t \mid X > u) = \mathrm{P}(X > t) = \mathrm{e}^{-\lambda t},$$

whatever u is. So A's *remaining* service time from the moment I walk in has the same exponential distribution with parameter λ, and the same goes for B. The two remaining times are independent because the full service times are. Hence

$$\mathrm{P}(\text{both still being served at time } t) = \mathrm{e}^{-\lambda t} \cdot \mathrm{e}^{-\lambda t} = \mathrm{e}^{-2\lambda t}.$$

Am I still being served when the other customer finishes?

Say A finishes first, at time τ after I arrive, and I go straight to A's assistant.



At time τ, let R be the time B still needs, and let M be my service time.

- M is exponential with parameter λ, since my service starts fresh at τ. It is independent of everything before, because service times for different customers are independent.
- R is also exponential with parameter λ. At time τ, B has already been served for some length of time (the time before I arrived plus τ), and by the memoryless property again the remaining time has the original distribution. It is independent of M.

I am still being served when B finishes exactly when $M > R$. Now M and R are independent with the *same* continuous distribution, so by symmetry $\mathrm{P}(M > R) = \mathrm{P}(R > M)$. Also $\mathrm{P}(M = R) = 0$ for continuous random variables, so these two probabilities add to 1. Hence

$$\mathrm{P}(\text{I am still being served when the other customer finishes}) = \tfrac{1}{2}.$$

If you prefer a calculation to the symmetry argument, condition on R:

$$\mathrm{P}(M > R) = \int_0^\infty \lambda \mathrm{e}^{-\lambda r}\, \mathrm{P}(M > r)\, \mathrm{d}r = \int_0^\infty \lambda \mathrm{e}^{-2\lambda r}\, \mathrm{d}r = \tfrac{1}{2}.$$

Checks

- At $t = 0$, $e^{-2\lambda t} = 1$: both customers are certainly being served when I arrive. As $t \to \infty$ the probability tends to 0, as it should.
- The answer $\frac{1}{2}$ doesn't depend on λ, which is what we'd expect, because changing λ only changes the unit of time.
- A simulation with $\lambda = 0.7$ and $t = 0.9$ gave 0.2831 for both customers still being served, against $e^{-1.26} \approx 0.2837$. In a simulation where both customers had already been in service for random lengths of time, I was still being served in a proportion 0.5000 of trials.

Common slips

- Thinking the customers already being served are "due to finish soon", so that their remaining times are shorter. For the exponential distribution, what has already happened makes no difference.
- Writing $\mathrm{P}(X > s + t \text{ and } X > t)$ as $\mathrm{P}(X > s + t)\,\mathrm{P}(X > t)$. These events are not independent; one contains the other.

Beyond STEP

The exponential is the only continuous distribution with the memoryless property. That fact underpins the *Poisson process* and *queueing theory*, where a shop with c assistants and exponential service times is the classic M/M/c queue. The calculation for two customers extends to n: the minimum of n independent exponential variables with parameter λ is exponential with parameter $n\lambda$ (multiply the tail probabilities), and by symmetry each variable is equally likely to be the smallest. Grimmett and Welsh, *Probability: An Introduction*, covers exponential variables and the Poisson process at this level.

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